Leakage Detection in Water-Distribution-Networks using Physics-Aware Neural Networks and Leakage Localization using a Rule-based System

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Abstract

Water Distribution Networks (WDNs) lose approximately 126 billion m³ annually to leaks, costing 39B USD. We propose a Physics-Aware Artificial Neural Network (PA-ANN) that embeds conservation of mass and the Hazen-Williams equation directly into the training objective. Evaluated on the LeakDB Hanoi dataset (17.5M data points), our approach achieves 96.59% precision (5.64% improvement over baseline) while reducing false positives by 62.98%. A novel topology-informed localization algorithm with anti-drift mechanism reduces fault isolation time by 63.6% while maintaining 79.17% localization accuracy. Results demonstrate that incorporating physical constraints improves both detection reliability and operational efficiency.

Keywords: keyword 1, keyword 2, keyword 3, keyword 4

1. Introduction

1.1. Motivation and Relevance

Water is one of the most essential resources for human survival, economic growth, and sustainable urban development. With rapid urbanization and increasing population density in major towns and cities, Water Distribution Networks (WDNs) have become more extensive and complex. These networks are responsible for delivering potable water from natural water bodies or reserviors to consumers, and their efficient operation is crucial to ensure water security and sustainability. The volume of leakages in WDNs varies throughout different geographies, and the primary factors affecting the amount of water lost due to leakage are the general condition of the network and the maintenance and monitoring system in place. According to Serafeim et al. (2024), Some places suffer from as much as 70-80% loss in water due to leakages, while places and areas where the infrastructure is secure, the lost water volume is as low as 7-8%. In total, around the World, the amount of non-revenue water (NRW) is around 126 billion cubic meters annually, and amounts to monetary loss of approximately 39 billion dollars(USD.)

1.2. Challenges

Monitoring systems for WDNs are primarily acoustic monitoring systems, or they rely heavily on manual inspection and are labor-intensive, time-consuming, and costly. Recent advancements in data-driven methods and sensor-based monitoring have improved detection accuracy; however, many of these approaches rely heavily on data driven and complex models that do not explicitly incorporate the underlying physical laws of fluid dynamics. As a result, their predictions may lack physical interpretability and generalization in unseen scenarios.

1.3. Contribution

Our model, which is Physics-Aware ANN (PAANN), tries to integrate physical laws into the model latent space so the predictions do not violate the equations and laws of conservation of mass and energy, and the models vastly reduce the number of false-positives which save both the monetary and human cost, predicting a leak only when the model is very sure. This is a comparatively low cost solution that can be generalized for any WDN, and our localization method predicts the leakage area using a relatively simple set of rules, not relying on expensive models for localization which are both time and compute consuming.

2. Related work

The detection and localization of leaks in Water Distribution Networks (WDNs) have been investigated through various methods. This section presents a synthesis of recent advances, organized into subsections showing the various methodologies applied, ranging from tradition data driven machine learning approaches to physics informed ones , along with the research gap motivating the present study.

2.1. Traditional Data-Driven Approaches

Classical machine learning and deep learning methods have shown considerable promise in leak detection, primarily by learning patterns from historical operational data. Mahdi et al. (2024) employed advanced statistical features—specifically, Autocorrelation Coefficient (Au-C) and Signal Energy (Sig-E)—extracted from accelerometer and dynamic pressure sensor measurements. Their Artificial Neural Network (ANN) achieved accuracies of 86.5% and 86.2% F1-scores for branched and looped configurations, respectively. The study highlighted the efficacy of data fusion in enhancing detection reliability; however, the approach

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remained purely empirical, lacking integration with underlying hydraulic principles.

Similarly, Sousa et al. (2023) investigated machine learning strategies including K-means clustering, Learning Vector Quantization (LVQ), and cluster validation techniques on data from a District Metered Area (DMA) in Stockholm, Sweden. Their methodology achieved classification rates up to 93.98%, demonstrating robustness to common sensor noise levels within ±5%. Nonetheless, the authors acknowledged limitations in handling dynamic operational variations and the absence of physical interpretability in model predictions.

Deep learning architectures have also been explored. Punukollu et al. (2023) applied Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks on the LeakDB benchmark dataset, achieving leak detection accuracies ranging from 49% to 98.23% depending on network complexity. While these models captured temporal dependencies effectively, they required extensive labeled training data and exhibited reduced generalization under distribution shifts which is a common challenge in purely data-driven paradigms.

2.2. Hybrid and Model-Based Methodologies

Recognizing the limitations of purely data-driven approaches, recent research has integrated hydraulic modeling with statistical or machine learning techniques. Mazaev et al. (2022) proposed a hybrid framework combining EPANET hydraulic simulations with a Time-Windowed Head Bias Correction (TWHBC) mechanism and an Elastic-Net logistic regression classifier. Their approach explicitly accounted for model-data discrepancies through bias correction and achieved leak localization within predefined search radii for multiple in-field experiments. However, the methodology's reliance on accurate hydraulic model calibration and computational overhead limited its real-time applicability.

Luo et al. (2024) propose a novel transformer based model for the efficient detection of leakages in WDNs. The authors utilize a transformer architecture with an attention mechanism designed to learn normal pressure distributions by evaluating correlations between immediate historical data and data from the exact same time on different days. A key methodological improvement in their approach is concatenating rather than directly adding position embeddings to the pressure data, which prevents the blurring of subtle anomalous features caused by leaks.

Barros et al. (2025) introduced a multilayer network representation integrating topological, correlation-based, and sensor coverage graphs. By combining graph theory with statistical anomaly detection (z-score and Interquartile Range), they successfully detected leaks within 15 minutes and localized them within a 50-meter range in the BattLeDIM benchmark network. Their approach reduced the need for exhaustive hydraulic simulations by regionalizing search spaces; however, the framework struggled with incipient leaks exhibiting gradual flow increases, underscoring the necessity for continuous physical constraint enforcement.

2.3. Physics-Informed Neural Networks in WDNs

Physics-Informed Neural Networks (PINNs) represent an emerging paradigm that embeds governing partial differential equations (PDEs) directly into neural network loss functions, ensuring adherence to known physical laws. Tornyeviadzi and Seidu (2023) pioneered the application of a 1D CNN deep autoencoder with semi-supervised learning on multivariate SCADA data from the L-TOWN benchmark network. Their PINN-based framework incorporated continuity and momentum equations via Pruned Exact Linear Time (PELT) change point detection, achieving identification of 16 out of 19 leaks and accurate localization of 13 within a 300-meter radius—without requiring a calibrated hydraulic model. The study demonstrated minimal identification lag for abrupt leaks, though incipient leaks remained challenging.

Expanding on this foundation, Falas et al. (2020) explored PINNs for cybersecurity applications in WDNs, specifically mitigating sensor-based cyberattacks. By training surrogate models on Navier-Stokes equations, they achieved predictive accuracies exceeding 99% for velocity fields, enabling the detection of data integrity attacks within simulated environments. While conceptually innovative, the work focused on small-scale proof-of-concept scenarios and did not address scalability to operational networks.

More recently, Lin et al. (2025) developed a digital twin framework leveraging PINNs for pipeline leakage detection in Taiwan's water infrastructure. Their model embedded continuity and momentum equations into the loss function and utilized statistical control charts (z-score and IQR) on reconstruction errors to detect anomalies. Validation on real-world datasets yielded R^2 values of 88.91% and 98.91% for small- and large-scale networks, respectively, with leak detection within 15 minutes of occurrence. Despite these successes, the framework's dependency on accurate boundary conditions and its limited handling of transient flow regimes were noted as areas requiring further investigation.

2.4. Benchmark Datasets and Evaluation Challenges

The availability of standardized datasets is crucial for reproducible research. Kyriakou (2018) introduced LeakDB, a publicly available benchmark comprising EPANET-simulated scenarios for multiple WDN topologies (including Hanoi, Net1) with annotated leak events. The dataset facilitates comparative evaluation but has been critiqued for relying solely on simulated data, potentially limiting model robustness under real-world operational uncertainties.

Comparative studies on BattLeDIM Barros et al. (2025) and other benchmarks reveal persistent challenges: high false positive rates for incipient leaks, computational overhead in large-scale networks, and difficulty in achieving physically interpretable predictions under domain shifts. These gaps motivate the integration of physics-informed constraints within scalable ANN architectures.

2.5. Research Gaps and Motivation

Despite substantial progress, existing methodologies exhibit critical limitations:

- **Lack of Physical Interpretability:** Purely data-driven models Mahdi et al. (2024); Punukollu et al. (2023) achieve high accuracies but fail to guarantee adherence to conservation laws, limiting generalization and trustworthiness.

- **Computational Scalability:** Hybrid approaches Mazaev et al. (2022) and traditional PINNs Lin et al. (2025) often require extensive hydraulic simulations or suffer from high computational costs in large networks.

- **Incipient Leak Detection:** Methods leveraging statistical anomaly detection Barros et al. (2025); Tornyeviadzi and Seidu (2023) struggle with gradual leak growth due to subtle signal variations.

Addressing these gaps, the present study proposes a **Physics-Aware Artificial Neural Network (PA-ANN)** framework that embeds conservation of mass, the Hazen-Williams relation directly into a Feed-Forward Neural Network architecture. By using the LeakDB Hanoi dataset, we aim to achieve physically consistent computationally efficient leak detection, advancing beyond existing data-driven and hybrid methodologies.

3. What is a Physics-Aware Neural Network?

3.1. Physics-Informed Neural Networks (PINNs)

PINNs are neural networks that solve ***continuous partial differential equations (PDEs)*** that govern physical systems. This is done by embedding differential equations into the model training. The derivatives are calculated using ***automatic differentiation.*** The loss is defined as the PDE residual, evaluated at points in the continuous domain.

3.2. Physics-Aware Neural Networks (PANNs)

In comparison, as our data is discrete (at 30 minutes time-step), we use PANNs, here the model predicts the physical quantity along with its goal, and the predicted values are used to calculate the residuals of mass and energy using some formulae that seem appropriate for the data. The residuals (normalized) are added to the data loss for backprop, and the predicted values play a role in the classification part of the model. **Compared to PINNs, PANNs take significantly less training time and compute resources for a significant increase in performance over baseline models.**

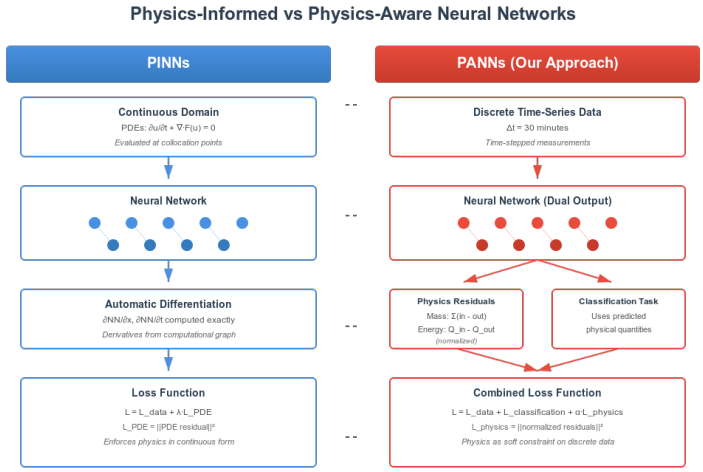

Figure 1: PINN vs PANN

4. Dataset

4.1. Dataset Description

This study has used the LeakDB Hanoi dataset, which is made using the *Water Network Resilience Tool (WNTR)*. The dataset consists of a total of 999 scenarios, where each scenario is a different simulation of the Hanoi WDN for a year. The data points in each scenario are thirty minutes apart, which means there are 17,520 data points in one scenario, and a total of 17,502,480 data points in the entire dataset.

The Hanoi WDN consists of 32 nodes and 34 pipes connecting them, and the LeakDB dataset assumes pressure, demand and flow-rate sensors at these 32 nodes. Thus, each data point has 32 demand readings, 32 pipe readings and 34 flow-rate readings, making a total of 98 values per time-step.

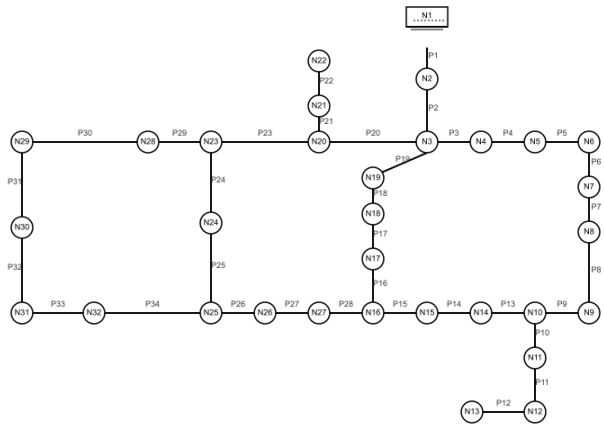

Figure 2: Figure of Hanoi WDN

The Data can be downloaded using the `water_benchmark_hub` Python module. There are two types of reading available, namely *raw_data* and *scada_data*. In this work, *raw_data* is used, as it also includes readings of demand.

Initially, each scenario is stored as a directory containing subfolders for *demand*, *pressure*, *flow*, and *leaks*, where leak labels are multi-class values ranging from 0 to 32. In addition, each scenario includes an *INP* file, which specifies the simulated length, diameter, and roughness coefficient of each pipe. For the learning task, the dataset is reformatted into a single `.csv` file per scenario, where leak labels are converted to a binary representation (leak / non-leak).

Leak localization is not learned directly by the model and is instead performed as a post-processing step using a rule-based system.

4.2. Exploratory Analysis of Data

Below is the heatmap of all 999 scenarios with 5% rows from each scenario selected randomly. The demand and pressure readings show a constant negative correlations, reflecting drop in pressure when demand increases or vice-versa. Flow readings show mixed correlations with demand and pressure, indicating the directional and topological effects of the network. These correlations support the use of dense models and also physics-aware models.

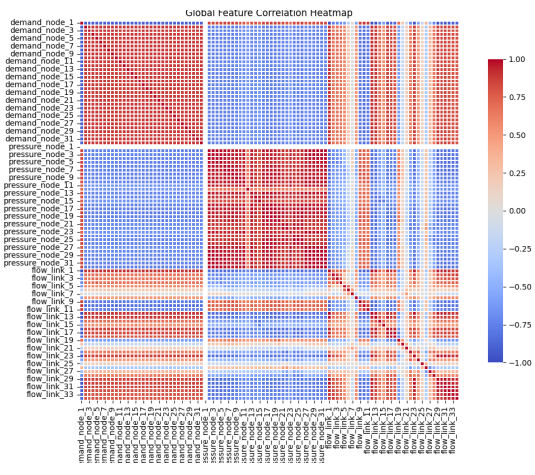

Figure 3: Correlation heatmap of Hanoi WDN

In the dataset, 43,11,716 rows have a leak state, while 1,31,90,764 rows a non-leak state, this means that that the ratio of non-leak data-points to leak points is **3.059**, making the dataset very imbalanced and a challenge for any neural network.

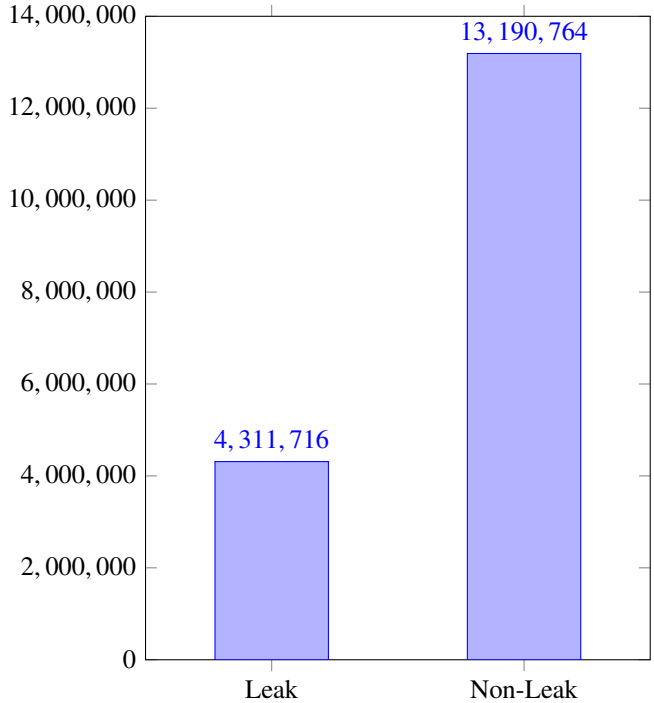

Figure 4: Distribution of leak and non-leak rows in the dataset.

A Scenario which has at least one leak state can be called a *leak scenario*, while a scenario which has no leak state in any timestamp can be called a *non_leak scenario*. Out of the 999 scenarios, 758 are *leak scenarios*, while 241 are *non_leak scenarios*. This ratio is also maintained while selecting the scenarios for training, validation, and testing.

4.3. Feature Engineering and Pre-Processing

In terms of feature engineering, two features have been added to each scenario for an increase in model performance. As the data does not have any missing or *NaN* values, filling in missing data is not needed and only data normalization has been performed in terms of pre-processing.

4.3.1. Adding columns

In the day time, due to usage of water both commercial and industrial, the general trend of pressure and flow is downward, while demand is higher and vice-versa during the night time. In order to prevent the models from reading the downward trend of pressure and flow and keep returning leak predictions, two columns, *sin_hour* and *cos_hour*, are added to each scenario, in order to add weight to the particular hour of the day when a leak occurs.

Let i denote row index of a scenario, Thus *step of day* $\boldsymbol{s}$ is defined as:

$$s_i = i \bmod 48$$

The temporal features are described as:

$$\text{sin_hour}_i = \sin\left(\frac{2\pi s_i}{48}\right)$$

$$\text{cos_hour}_i = \cos\left(\frac{2\pi s_i}{48}\right)$$

Because of these columns, a time step of 12 A.M. (sin_hour = 0 and cos_hour = 1) is given more bias by the model for a positive classification than 9 A.M. (sin_hour = 0.707 and cos_hour = -0.707) for similar readings of pressure, demand and flow.

4.3.2. Normalization

All sets (train, validation, and test) are normalized using the mean and standard deviation of the training set, such that the mean of each set is 0 and the standard deviation is 1, after normalization.

Let $\mathcal{D}_{\text{train}} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$ denote the training dataset, where each feature vector $\mathbf{x}_i \in \mathbb{R}^F$.

The global mean and standard deviation are computed feature-wise over the training set as:

$$\boldsymbol{\mu} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i, \tag{1}$$

$$\boldsymbol{\sigma} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i^{\circ 2} - \boldsymbol{\mu}^{\circ 2}}, \tag{2}$$

where $\circ$ denotes element-wise operations.

To ensure numerical stability, the variance is lower bounded by a small constant ϵ:

$$\boldsymbol{\sigma} = \sqrt{\max\left(\frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i^{\circ 2} - \boldsymbol{\mu}^{\circ 2}, \epsilon\right)}. \tag{3}$$

Using the statistics computed from the training set, each feature is normalized as:

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \tag{4}$$

where $\tilde{\mathbf{x}}$ denotes the normalized feature vector.

The same training-set mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$ are applied consistently to the training, validation, and test sets to avoid data leakage.

4.4. Windowing function

To give the model temporal context and features without the use of recurrent architectures such as LSTMs or GRUs, a sliding window based dataset construction is used. Temporal windows are extracted from each scenario using a fixed window size of 12, or 6 hours as each time-step is of 30 minutes. For each window, the target is the leak state (0/1) at the final time stamp of the window, allowing the model to learn temporal patterns preceding a leak event. As the target label is the leak state at the final time stamp, which is also the current time stamp, this means that the model **nowcasts** leakage prediction.

4.4.1. ANN

For the ANN models, each window is flattened into a one-dimensional vector as input to the network. A window of 12 time stamps with 100 features per time stamp is reshaped into a vector of dimension one and length 12×100. The mathematical representation is:

Let a scenario be represented as a multivariate time series

$$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}, \quad \mathbf{x}_t \in \mathbb{R}^F, \tag{5}$$

where F denotes the number of features at each time step.

Using a sliding window of length W, the k-th temporal window is defined as

$$\mathbf{X}^{(k)} = [\mathbf{x}_k, \mathbf{x}_{k+1}, \ldots, \mathbf{x}_{k+W-1}] \in \mathbb{R}^{W \times F}. \tag{6}$$

For the ANN model, each window is reshaped into a one-dimensional feature vector via vectorization:

$$\tilde{\mathbf{x}}^{(k)} = \text{vec}\left(\mathbf{X}^{(k)}\right) \in \mathbb{R}^{W \cdot F}. \tag{7}$$

This flattened representation serves as the input to the fully connected network.

5. Physics Loss

All physical systems are governed by the conservation of mass and conservation of energy, and the Water Distribution Network is also assumed to be governed by these laws. The network is modeled as a directed graph, where vertices represent nodes, and edges represent pipes connecting nodes. *Mass conservation* is enforced at each node, and *Energy conservation* is enforced along each pipe. The equations used for these are discussed below.

5.1. Equations

For *Mass conservation*, the **Continuity Equation** is used. This means that the inflow at each node is equal to the sum of the demand at that node and the outflow at each node.

$$r_{\text{mass}}^{(i)} = \sum_{j \in \mathcal{E}_{\text{in}}(i)} Q_j - \sum_{j \in \mathcal{E}_{\text{out}}(i)} Q_j - D_i,$$

where Q_j denotes the flow through pipe j, D_i is the demand, and $\mathcal{E}_{\text{in}}(i)$ and $\mathcal{E}_{\text{out}}(i)$ represent incoming and outgoing pipes, respectively.

For *Energy conservation*, the *Hazen-Williams Equation* is used. The **Hazen-Williams Equation** defines the loss due to friction in a pipe; this frictional loss, along with the elevation head differences between the two ends of a pipe, gives the pressure drop. Since in the LeakDB dataset, in all scenarios, all nodes are kept at the same elevation, the pressure drop due to elevation differences is assumed to be negligible. The Hazen-Williams head loss for pipe k is given by

$$h_k = 10.67 \frac{L_k |Q_k|^{1.852}}{C_k^{1.852} D_k^{4.87}},$$

where L_k is the pipe length, D_k is the diameter, C_k is the roughness coefficient, and Q_k is the predicted flow.

5.2. Physics Loss Algorithm

The physics loss due to mass conservation is defined as:

$$\mathcal{L}_{\text{mass}} = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{r_{\text{mass}}^{(i)}}{\bar{D}} \right)^2,$$

where $\bar{D}$ denotes a scale factor (e.g., mean absolute demand) for numerical stability.

The predicted pressure drop across a pipe is given as:

$$\Delta p_k = \text{sign}(Q_k)\,(p_{\text{src}(k)} - p_{\text{dst}(k)}),$$

where $p_{\text{src}(k)}$ and $p_{\text{dst}(k)}$ are the predicted pressures at the source and destination nodes.

Thus, the loss due to energy conservation is defined as:

$$r_{\text{energy}}^{(k)} = \Delta p_k - h_k,$$

$$\mathcal{L}_{\text{energy}} = \frac{1}{M} \sum_{k=1}^{M} \left(\frac{r_{\text{energy}}^{(k)}}{\bar{p}} \right)^2,$$

where $\bar{p}$ denotes a scale factor (e.g., mean absolute predicted pressure) for numerical stability.

The total physics loss is a weighted average of the loss due to mass conservation and energy conservation, given as:

$$\mathcal{L}_{\text{phys}} = \lambda_{\text{mass}} \mathcal{L}_{\text{mass}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}},$$

where λ_{mass} and λ_{energy} are hyperparameters; in our implementation we use $\lambda_{\text{mass}} = 1.0$ and $\lambda_{\text{energy}} = 0.1$ to emphasize mass conservation consistency.

5.3. Pseudocode and Training Integration

Algorithm 1 Physics Loss Computation

Require: $\hat{p}$ (pred. pressure), $\hat{Q}$ (pred. flow), D (demand), L, D, C (pipe params), edge_index
Ensure: $\mathcal{L}_{\text{phys}}, \mathcal{L}_{\text{total}}$

Aggregate Flow at Nodes
$\text{flow_in}[i] \leftarrow \sum_{j:\text{dst}(j)=i} \hat{Q}_j$
$\text{flow_out}[i] \leftarrow \sum_{j:\text{src}(j)=i} \hat{Q}_j$
Mass Conservation Residual
$r_{\text{mass}}^{(i)} \leftarrow \text{flow_in}[i] - \text{flow_out}[i] - D_i$
$\mathcal{L}_{\text{mass}} \leftarrow \text{mean}\left(\left(\frac{r_{\text{mass}}}{\bar{D}}\right)^2\right)$
Hazen–Williams Head Loss
$h_k \leftarrow 10.67 \cdot L_k \frac{|\hat{Q}_k|^{1.852}}{C_k^{1.852} D_k^{4.87}}$
Predicted Pressure Drop
$\Delta p_k \leftarrow \text{sign}(\hat{Q}_k)(\hat{p}_{\text{src}(k)} - \hat{p}_{\text{dst}(k)})$
Energy Residual
$r_{\text{energy}}^{(k)} \leftarrow \Delta p_k - h_k$
$\mathcal{L}_{\text{energy}} \leftarrow \text{mean}\left(\left(\frac{r_{\text{energy}}}{\bar{p}}\right)^2\right)$
Total Physics Loss
$\mathcal{L}_{\text{phys}} \leftarrow \lambda_{\text{mass}} \mathcal{L}_{\text{mass}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}}$
Combined Training Loss
$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{WBCE}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}$

Algorithm 2 Training Loop with Physics Loss

Require: Training data $(x_b, y_b, \text{scenario_ids}, D_b)$
Ensure: Updated model parameters

Set Physics Weight by Epoch
if $e < 10$ **then**
  $\lambda_{\text{phys}} \leftarrow 0.01$
else if $10 \le e < 20$ **then**
  $\lambda_{\text{phys}} \leftarrow 0.03$
else
  $\lambda_{\text{phys}} \leftarrow 0.05$
end if
for each batch $(x_b, y_b, \text{scenario_ids}, D_b)$ **do**
  Load pipe parameters (L, D, C) for batch
  Denormalize D_b to physical units
  $(\hat{y}, \hat{p}, \hat{Q}) \leftarrow \text{model}(x_b)$
  Denormalize $\hat{p}, \hat{Q}$
  $\mathcal{L}_{\text{WBCE}} \leftarrow \text{weighted_BCE}(\hat{y}, y_b)$
  Compute $\mathcal{L}_{\text{total}}$ using Algorithm 1
  Backpropagate $\mathcal{L}_{\text{total}}$
  Update model parameters
end for

5.4. Integration into Model

To use this loss in the ANN model, the *.inp* files of each scenario are formatted into a *.csv* file containing the pipe length, diameter, and roughness coefficient of each pipe in the scenario, and this information is also integrated into the dataset class. As the node-pipe connection is constant for all scenarios, a constant *edge_list* is made and used. While training, the physics loss is added with the classification loss, and the result is used for backpropagation.

5.5. The Thought

Our idea with this loss was to constrain the model's latent space for such predictions that were physically possible and did not violate any of the equations, and that gradually the residual due to the difference between the true pressure and flow, and the predicted pressure and flow by the physics-aware models kept going closer and closer to 0. The predicted values of pressure and flow are also used in classification, discussed in more detail in the model architecture.

6. Model Architecture and Training Details

6.1. Architecture

6.1.1. Base ANN Model

The model here is a fully connected feed-forward network designed for classification of presence of leak on a single timestamp. The input layer consists of 1200 features, due to the Windowing function. It is passed through three hidden layers of 512, 256 and 64 neurons. The first two hidden layers employ Batch Normalization and ReLU activation, and a Dropout Layer (rate = 0.5) to prevent overfitting. The third hidden layer uses only ReLU activation. The output layer consists of a single neuron with a sigmoid activation function, for binary leak classification.

ANN Loss:

As discussed above, because the model is a binary classification model, and the class imbalance is severe, **Class-weighted binary cross-entropy** loss is used, where the loss is reweighted inversely proportional to class frequency.

Let N_0 denote the number of non-leak samples, and let N_1 denote the number of leak samples.

$$w_0 = \frac{N_0 + N_1}{2N_0}, \qquad w_1 = \frac{N_0 + N_1}{2N_1}$$

Let $y_i \in \{0, 1\}$ denote the ground-truth leak label for the i-th sample, and let $\hat{y}_i \in (0, 1)$ denote the corresponding predicted leak probability.

$$\mathcal{L}_{\text{WBCE}} = -\frac{1}{N} \sum_{i=1}^{N} \left[w_1 \, y_i \log(\hat{y}_i) + w_0 \, (1 - y_i) \log(1 - \hat{y}_i) \right]$$

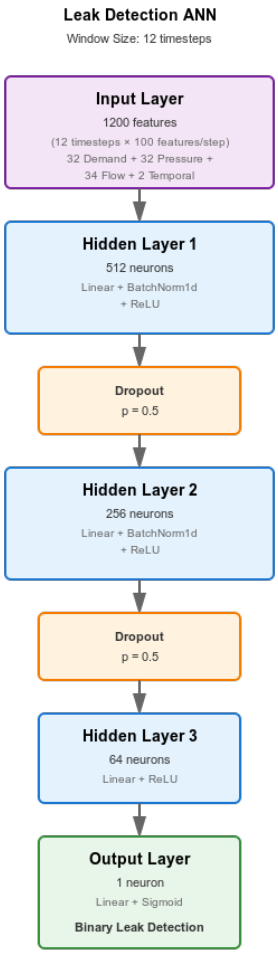

Figure 5: Architecture Diagram of ANN

6.1.2. *Physics-Aware ANN Model*

The model here is a fully connected feed-forward network designed for prediction of pressure and flow, and classficition of presence of leak on a single timestamp, using those predictions. The input layer is 1200 features here too. It is passed through three hidden layers of 512, 256 and 64 neurons. All three hiden layers employ Batch Normalization, ReLU Activation and a Dropout Layer (rate = 0.5). The third hidden layer is connected to 2 layers, one with 32 neurons, and one with 34 neurons, to predict pressure and flow, respectively. The Leak head is consists of 2 layers, with the input layer being the 64 neurons of the third hidden layer, and 5 features that are derived from the predicted pressure and flow (mean of pressure and flow, standard deviation of pressure and flow, and the range of pressure), thus 69 features in total, which are then passed through a hidden layer of 64 neurons, where ReLU Activation and Dropout Layer is employed, and the output layer is a single neuron with a sigmoid activation function, for binary leak classification.

Physics-Aware ANN Loss:

The **Class-weighted binary cross-entropy**, also the **Classification loss** of the model is given by:

Let N_0 denote the number of non-leak samples, and let N_1 denote the number of leak samples.

$$w_0 = \frac{N_0 + N_1}{2N_0}, \qquad w_1 = \frac{N_0 + N_1}{2N_1}$$

Let $y_i \in \{0, 1\}$ denote the ground-truth leak label for the i-th sample, and let $\hat{y}_i \in (0, 1)$ denote the corresponding predicted leak probability.

$$\mathcal{L}_{\text{WBCE}} = -\frac{1}{N} \sum_{i=1}^{N} \left[w_1 \, y_i \log(\hat{y}_i) + w_0 \, (1 - y_i) \log(1 - \hat{y}_i) \right]$$

Let the physics loss, as discussed above in detail above, be represented as: $\mathcal{L}_{\text{phys}}$

Thus, the total loss of the model is given by:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{WBCE}} + \lambda_{\text{phys}} \, \mathcal{L}_{\text{phys}}.$$

The λ_{phys} follows this schedule:

$$\lambda_{\text{phys}}(e) = \begin{cases} 0.01, & e < 10, \\ 0.03, & 10 \le e < 20, \\ 0.05, & 20 \le e < 30, \end{cases}$$

This is done to make sure that the physics loss does not dominate the classification loss at the start.

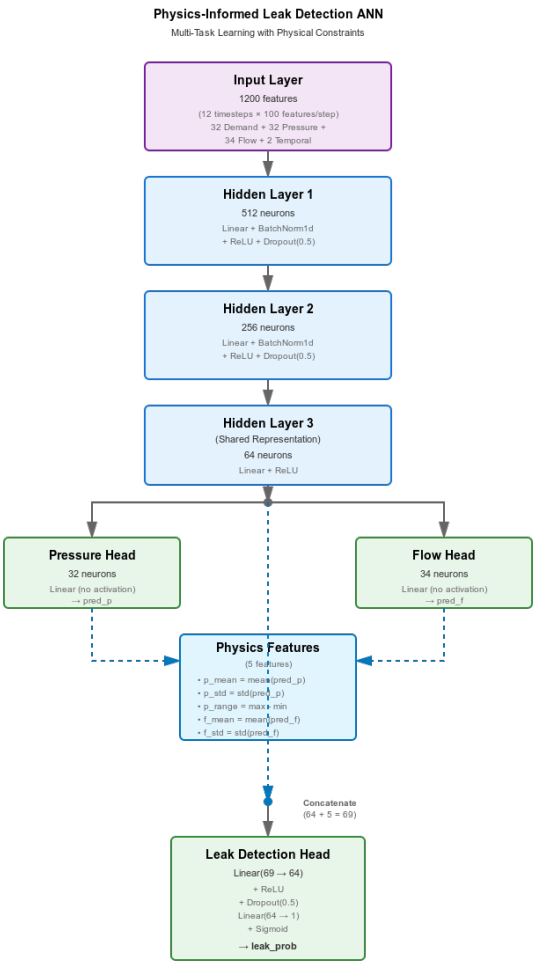

Figure 6: Architecture Diagram of Physics-Aware ANN

7. Localization

7.1. Introduction

While the Deep Learning (ANN) module handles the temporal detection of anomalies (identifying *when* a leak occurs), spatial localization (identifying *where* the leak is located) is managed by a topology-informed statistical engine. Designed specifically for real-time operation, this algorithm processes sequential data streams. This hybrid architecture leverages the strengths of both domains: the ANN captures complex, non-linear patterns to detect incipient events, while the localization module relies on hydraulic first principles and network graph theory.

A major challenge in hydraulic localization is signal drift. When a leak occurs, the initial pressure drop is most severe at the source node. However, as the transient wave propagates, pressure drops accumulate at physical boundaries, such as dead-ends or downstream junctions, often causing purely statistical algorithms to drift away from the true source over time. To combat this, the proposed real-time module introduces an "Anti-Drift" mechanism combining time-decayed voting, topology-based neighbor validation, and a stability lock, triggered only when the ANN confidence exceeds a threshold ($\tau_{conf} > 0.75$).

7.2. Algorithm

The real-time localization algorithm operates in three distinct phases: Lagged Baseline Snapshot, Topological Scoring, and Time-Decayed Voting.

7.2.1. 1. Lagged Baseline Snapshot

During normal operation (when the ANN probability $P(leak) < 0.60$), the system maintains a lightweight rolling buffer of pressure readings for all N sensor nodes. Upon the ANN triggering a leak state at time $t_{trigger}$, the system immediately captures a "snapshot" of the network's healthy state.

To prevent the baseline from being contaminated by the incipient, pre-trigger transient wave of the leak, a lag period ($W_{lag} = 50$ steps) is excluded. The reference mean (μ_i) and standard deviation (σ_i) for each sensor i are calculated from a stable historical window ($W_{base} = 200$ steps) preceding the lag:

$$\mathcal{B}_{clean} = P_{t_{trigger}-W_{lag}-W_{base} \,:\, t_{trigger}-W_{lag}} \tag{8}$$

This frozen baseline acts as the static reference frame for the duration of the event.

7.2.2. 2. Topological and Physics-Aware Scoring

At each active leak step t, the algorithm calculates the positive pressure residual and standardizes it into a Z-score, applying a noise floor ($\epsilon = 0.05$):

$$Z_i(t) = \frac{\max(0, \mu_i - p_i(t))}{\max(\sigma_i, \epsilon)} \tag{9}$$

Rather than evaluating nodes in isolation, the system utilizes the network's adjacency graph. A node i is evaluated based on its own drop, the supporting drops of its physical neighbors $\mathcal{N}_i$, and its structural degree D_i.

If $Z_i(t) > 1.0$, a Topological Score $S_i(t)$ is computed. It incorporates a Neighbor Support Ratio (R_i), rewarding nodes whose adjacent neighbors also exhibit drops, and a Degree Multiplier ($M_i = 1 + 0.05 \times D_i$), which lightly penalizes dead-ends ($D = 1$) in favor of central junctions:

$$R_i = \min\left(1.0, \frac{\frac{1}{|\mathcal{N}_i|}\sum_{j\in\mathcal{N}_i} Z_j(t)}{Z_i(t)}\right) \tag{10}$$

$$S_i(t) = Z_i(t) \times (1 + R_i) \times M_i \tag{11}$$

7.2.3. 3. Time-Decayed Voting and Anti-Drift Lock

To prioritize the initial hydraulic wave closest to the true leak source, the system applies a rapid time-decaying weight $W(t)$ based on the duration $\Delta t = t - t_{trigger}$:

$$W(t) = \frac{1}{1 + 0.1 \times \Delta t} \tag{12}$$

At each step, the top 3 candidate nodes are awarded points scaled by $W(t)$, and the cumulative vote V_i for each node is recursively updated.

To definitively prevent late-stage drift toward network boundaries, a **Stability Lock** is enforced. If the leading suspect $\hat{L}$ remains unchanged for a consecutive threshold of $k_{lock} = 10$ time steps, the algorithm considers the localization converged and halts suspect updating.

Handling Simultaneous Leaks: In scenarios involving multiple simultaneous bursts, the combination of time-decayed voting and the Stability Lock acts as a dominant-signal isolator. The node experiencing the most severe and immediate pressure gradient (the primary leak) accumulates early votes rapidly. The algorithm intentionally locks onto this primary source, preventing the system from oscillating confusingly between multiple coordinates. Secondary leaks can subsequently be localized once the primary anomaly is isolated by network operators and the baseline resets.

Algorithm 3 Real-Time Physics-Aware Localization Strategy

Require: Real-time Pressure stream P_t, ANN Model $\mathcal{M}$, Adjacency Graph $\mathcal{G}$
Ensure: Localized Node L_{est}

```
B ← ∅, V ← 0⃗, LeakActive ← False, IsLocked ← False
L_curr ← null, Stability ← 0
while stream is active at time t do
    p_leak ← M(P_{t−w:t})
    if not LeakActive then
        if p_leak < 0.60 then Append P_t to B
        end if
        if p_leak > 0.75 then
            LeakActive ← True, t_start ← t
            μ, σ ← ExtractLaggedStats(B, lag = 50, win = 200)
        end if
    else
        if p_leak < 0.40 then Reset System State
        end if
        if not IsLocked then
            W_t ← 1/(1 + 0.1 × (t − t_start))
            C ← ∅                                   ▷ Candidate List
            for each node i ∈ {1, ..., N} do
                Z_i ← max(0, μ_i − P_{i,t})/σ_i
                if Z_i > 1.0 then
                    Add (i, CalcTopoScore(Z_i, G)) to C
                end if
            end for
            Sort C descending by TopoScore
            for rank r ∈ {0, 1, 2} from top 3 of C do
                V_{C[r].id} ← V_{C[r].id} + (3 − r) × W_t
            end for
            L_new ← argmax(V)
            if L_new == L_curr then
                Stability ← Stability + 1
                if Stability > 10 then IsLocked ← True
                end if
            else
                L_curr ← L_new, Stability ← 0
            end if
        end if
    end if
end while
```

8. Results and Discussion

8.1. Metrics

We use four standard metrics: **Accuracy**, **Precision**, **Recall**, and **F1-Score** to evaluate the performance of our classification models.

Accuracy measures the overall ratio of correctly classified instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

$$\tag{14}$$

Precision measures how many positive predictions are also true positives.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{15}$$

$$\tag{16}$$

Recall measures how many of the true positives are actually predicted as true.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{17}$$

$$\tag{18}$$

F1-Score is a harmonic mean of the Precision and Recall, acting as a single number that tells us about the performance of any model.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{19}$$

8.2. Base Model Performance

The ANN model is trained on 490 scenarios, with the validation and test sets being 105 scenarios each. The ANN model is trained for a set 30 epochs, and the model with the highest validation F1 is used for testing. The model is tested at a set threshold of **0.5**.

Highest Validation F1 for the ANN model is **0.7483**.

Model Metrics:

Table 1: ANN Model Performance Metrics

Accuracy	Precision	Recall	F1
0.9213	0.9143	0.7889	0.8470

Confusion Matrix:

Table 2: ANN Confusion Matrix

	Predicted No Leak	Predicted Leak
Actual No Leak	1293741	37515
Actual Leak	107085	400104

Plot of Val F1 vs Epochs:

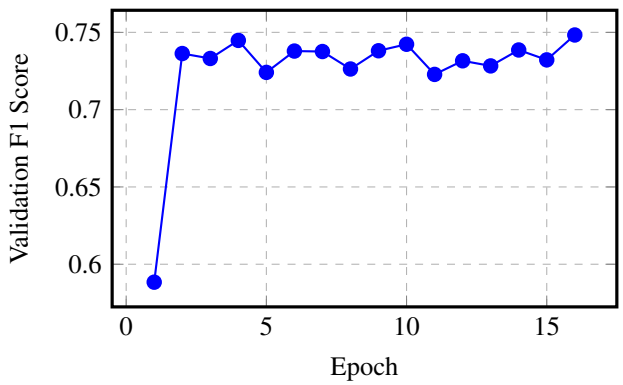

Figure 7: Validation F1 score across training epochs for ANN

8.3. Physics-Aware Model Performance

The ANN model is trained on 490 scenarios, with the validation and test sets being 105 scenarios each. The ANN model is trained for a set 30 epochs, and the model with the highest validation F1 is used for testing. The model is tested at a set threshold of **0.5**.

Highest Validation F1 for the ANN model is **0.7549**.

Model Metrics:

Table 3: Physics-Aware ANN Model Performance Metrics

Accuracy	Precision	Recall	F1
0.9303	0.9659	0.7747	0.8598

Confusion Matrix:

Table 4: Physics-Aware ANN Confusion Matrix

	Predicted No Leak	Predicted Leak
Actual No Leak	1317369	13887
Actual Leak	114272	392917

Plot of Val F1 vs Epochs:

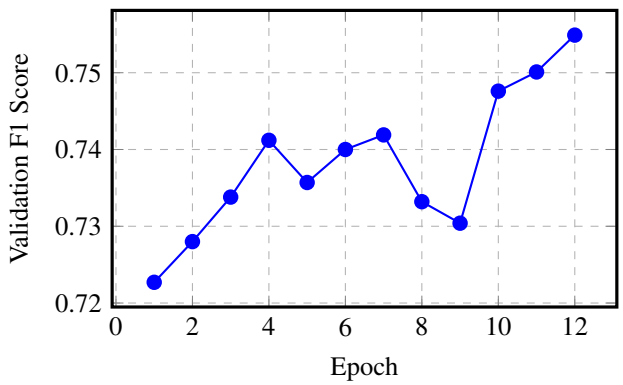

Figure 8: Validation F1 score across training epochs for Physics-Aware ANN

8.4. Comparison between ANN and Physics-Aware ANN

Table 5: Comparison between ANN and Physics-Aware ANN

Model	Accuracy	Precision	Recall	F1
ANN	0.92	0.91	**0.79**	0.85
Physics-Aware ANN	**0.93**	**0.97**	0.77	**0.86**

The Physics-Aware ANN achieves an improvement of 0.98% in accuracy, 5.64% in precision, and 1.51% in F1 compared to the ANN, while having a decrease of 1.79% in recall. The Physics-Aware ANN reduces the number of false positives by 23,628 (62.98% reduction), at the cost of increasing the false negatives by 7,187 (6.71% increase). This indicates that the Physics-Aware ANN is more conservative while predicting leaks, issuing leak predictions only when they are physically consistent and more confident than the ANN, which has a more liberal behavior while predicting leaks.

In practical deployment, this trade-off means that the Physics-Aware ANN incurs a slight delay in detecting certain leak events, however, it also drastically lowers the number of false alarms, resulting in a significant reduction of un-necessary human intervention and monetary costs.

8.5. Comparison with Existing Benchmarks on LeakDB Hanoi

To contextualize our results, we compare our approach with other methods evaluated on the LeakDB Hanoi dataset. Punukollu et al. (2023) applied Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks on the LeakDB benchmark dataset, achieving leak detection accuracies ranging from 87% to 96% depending on the type of leak. However, their results do not include specific precision and recall metrics

Experiments on transformer models by Luo et al. (2024) significantly outperform traditional machine learning approaches by using daily periodicity and long-term global information. On the Hanoi network, it achieved an Accuracy of 95.19% and an F1-score of 97.52%. Additionally, fully data-driven anomaly detection approaches, such as the Convolutional Autoencoder proposed by Leonzio et al. (2023), use reconstructed pressure matrices to identify leaks, achieving an overall accuracy of 89% on the LeakDB dataset.

Table 6: Comparison of methods on LeakDB Hanoi dataset

Method	Accuracy	Precision	Recall	F1
Transformer	95.19%	-	-	97.52%
Autoencoder	89.00%	-	-	78.84%
CNN	88.86%–96.55%*	-	-	-
LSTM	87.21%–89.65%*	-	-	-
ANN	92.13%	91.43%	78.89%	84.70%
PAANN	**93.03%**	**96.59%**	77.47%	**85.98%**

*Averaged across mixed, incipient, and abrupt leak scenarios.

Our Physics-Aware ANN achieves superior precision (96.59%) compared to existing deep learning benchmarks, which is particularly critical for minimizing false alarms in operational leak detection systems. Importantly, this improvement is achieved with a substantially more parameter-efficient architecture, whereas conventional CNN and LSTM models typically contain on the order of ∼1–2 million trainable parameters, and Transformer-based architectures often require even higher parameter counts due to multi-head attention mechanisms. While the recall is slightly lower than some reported values, the markedly higher precision—combined with a compact model size—highlights the effectiveness of embedding physical constraints directly into the learning framework. By enforcing hydraulic consistency within the prediction process, the proposed approach reduces spurious detections, limits unnecessary operational interventions, and provides a computationally efficient yet hydraulically reliable solution suitable for real-world deployment.

Table 7: Model parameter comparison

Method	Trainable Parameters
Transformer	∼8–12 Million (estimated)
CNN-LSTM	∼1.0–1.2 Million (estimated)
ANN (Ours)	764,289
PAANN (Ours)	773,187

8.6. Localization Results

Table 8: Fault localization performance comparison

Model	True Positives (TP)	False Negatives (FN)
Physics-Aware ANN	95	25
Standard ANN	95	25
Autoencoder	-	-

Total fault cases:

$$\text{Total} = TP + FN = 95 + 25 = 120$$

Localization accuracy:

$$\text{Accuracy} = \frac{TP}{TP + FN} = \frac{95}{120} = 79.17\%$$

Both models achieve identical localization accuracy of 79.17%, indicating equivalent fault detection capability under the rule-based localization framework.

Time reduction achieved by the Physics-Aware ANN:

$$\Delta T = 31h\,19m - 11h\,24m = 19h\,55m$$

Converting to hours:

$$11h\,24m = 11.40 \text{ hours}, \quad 31h\,19m = 31.32 \text{ hours}$$

Percentage reduction in localization time:

$$\text{Time Improvement} = \frac{31.32 - 11.40}{31.32} \times 100 \approx 63.6\%$$

Fault localization in this study is performed using a deterministic rule-based decision system applied uniformly to the outputs of both neural network models. The localization algorithm and implementation remain identical across experiments, ensuring that any observed performance differences arise solely from the predictive characteristics of the underlying models.

The Physics-Aware ANN incorporates physical constraints during training by optimizing a composite objective:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda \mathcal{L}_{physics}$$

where $\mathcal{L}_{data}$ represents the prediction error and $\mathcal{L}_{physics}$ enforces system-consistent physical relationships.

In contrast, the standard ANN minimizes only:

$$\mathcal{L}_{standard} = \mathcal{L}_{data}$$

Since the localization mechanism is rule-based and identical for both models, the equal accuracy (79.17%) indicates comparable prediction correctness. However, the Physics-Aware ANN reduces the average localization time by approximately 63.6%. This suggests that incorporating physical constraints produces outputs that align more efficiently with the rule-based decision logic, reducing ambiguity and accelerating fault isolation without modifying the localization algorithm itself.

8.7. Plots of Selected Scenarios

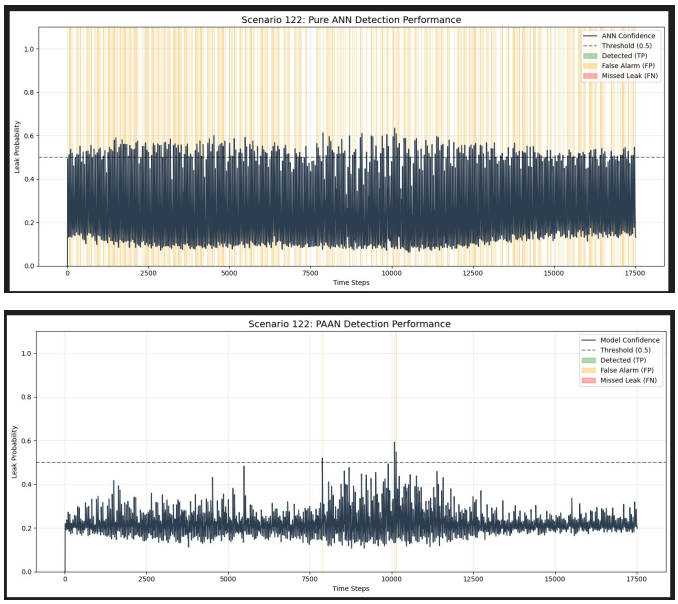

Figure 9: Scenario 122: ANN (top) and Physics-Aware ANN (bottom) scenario-wise detection.

The base ANN keeps generating high-frequency false alarms across the timeline, with confidence values frequently approaching the threshold, indicating poor discrimination under scenario 122. In contrast, the physics-aware ANN maintains stable low-confidence predictions, effectively eliminating false positives and improving reliability.

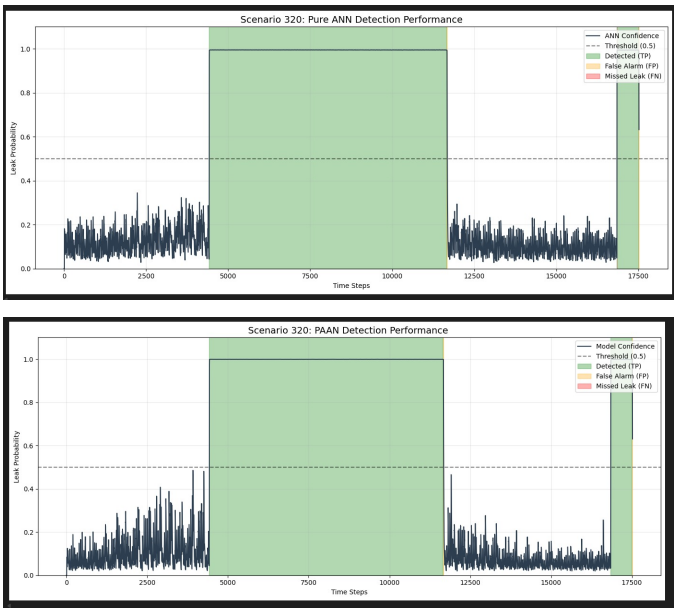

Figure 10: Scenario 320: ANN (top) and Physics-Aware ANN (bottom) scenario-wise detection.

The base ANN successfully detects the major leak intervals but shows sharp confidence transitions and fluctuations, indicating senstivity. The physics-aware ANN achieves clean and stable detection across the leak duration and minimal post leak fluctuations, demonstrating temporal consistency and reduced sensitivity.

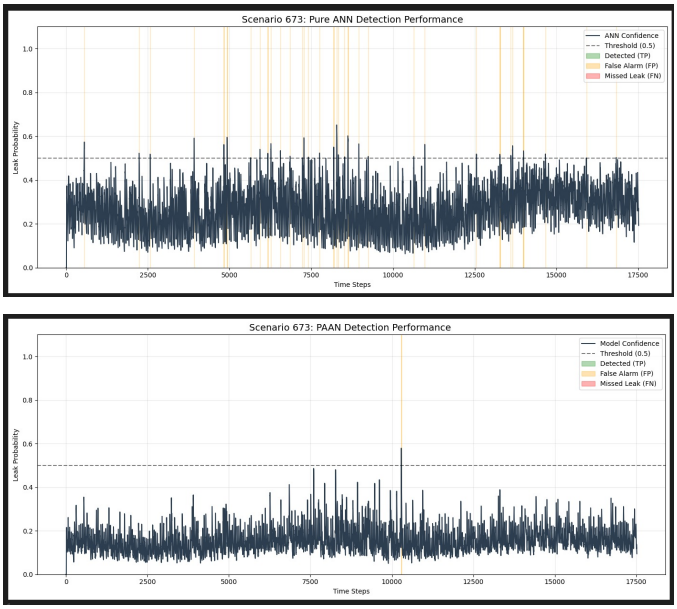

Figure 11: Scenario 673: ANN (top) and Physics-Aware ANN (bottom) scenario-wise detection.

The base ANN generates fluctuating confidence values around the decision threshold, resulting in false positives and unstable detections. The physics-aware ANN model exhibits significantly reduced confidence variance, with predictions remaining well be-

low the threshold and therefore eliminating spurious detections, indicating improved robustness under non-leak conditions.

9. Why Classical Machine Learning Techniques Are Not Used

To see if the detection problem can be solved using classical machine learning techniques, an XGBoost classifier was trained on the same windowed dataset. Table 9 presents the results for XGBoost classifiers with N-estimators = 50 and 100, tested at threshold 0.5 and at the optimal threshold derived by ROC-AUC (approximately 0.44 for both). While XGBoost performs decently, it lags behind both the base ANN and the Physics-Aware ANN. We can also see that the higher false negative rates of the classifiers are due to them being purely data-driven, and hence not gauranteeing conservation of mass and energy. Thus, this makes the ANN models, especially the Physics-Aware ANN suitable for deployment due to its embedding of hydraulic constraints.

Table 9: XGBoost on LeakDB Hanoi. N: trees; Thr.: threshold. ROC-AUC: 0.8705 (N=50), 0.8752 (N=100).

N	Thr.	Acc.	Prec.	Rec.	F1
50	0.50	0.8959	0.9519	0.6557	0.7765
50	0.44	0.8183	0.8695	0.7003	0.7758
100	0.50	0.9042	0.9567	0.6836	0.7974
100	0.44	0.8249	0.8905	0.7141	0.7926

10. Validity of Loss Function

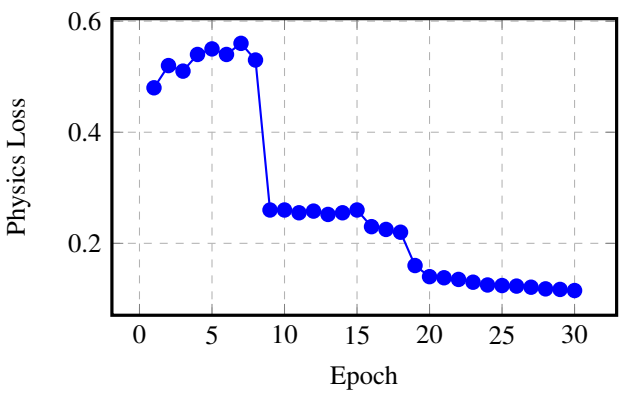

Figure 12: Physics loss convergence across training epochs

Any valid loss function used in optimization of a model should exhibit a decreasing trend during training if the model is able to learn the objective. Hence, similarly to how the weighted BCE Loss decreases across epochs as the model learns classification, the physics loss of the ANN model also decreases as it learns the underlying physics and the predicted pressure and flow values of the model leave smaller residuals across epochs.

11. Sensitivity Analysis

To assess model performance with change in Demand, Pressure and Flow, we perform a two-dimensional sensitivity analysis over demand pertubation $\alpha \in [-0.15, 0.15]$ and pressure-demand coupling strength $\beta \in [0.2, 1.0]$.

The formulae are:

$$\text{Demand}_{\text{new}} = \text{Demand}(1 + \alpha)$$

$$\text{Flow}_{\text{new}} = \text{Flow}(1 + \alpha)$$

$$\text{Pressure}_{\text{new}} = \text{Pressure}(1 - \beta\alpha)$$

These formulae are used keeping in mind the corelations shown by the data heatmap.

Figure 14 shows the worst case F1 (minimum over all β), and we can see that for negative demand shifts, the ANN degrades much more steeply as compared to the physics-aware ANN. At $\alpha = -0.15$, the ANN drops to an F1 of 0.17 whereas the physics-aware ANN holds at 0.31. For positive demand shift, both models show similar performance and the metrics are almost constant for $\alpha \in [0.05, 0.15]$.

Figure 15 shows the performance of both models at an extreme demand shift ($\alpha = 0.15$), for varying β. As we can see, the physics-aware ANN degrades gradually here, whereas the base ANN drops steeply from $\beta = 0.4$, where the F1 for both is approx 0.7, but at $\beta = 0.6$, the physics-aware ANN shows an F1 of 0.6 (approx), but the base ANN drops to an F1 of 0.4 (approx.).

As the physics-aware ANN embeds hydraulic constraints during training, even under data shifts that may not strictly follow conservation of mass and energy, it maintains more stable predictions, whereas the base ANN degrades drastically due to being purely data-driven.

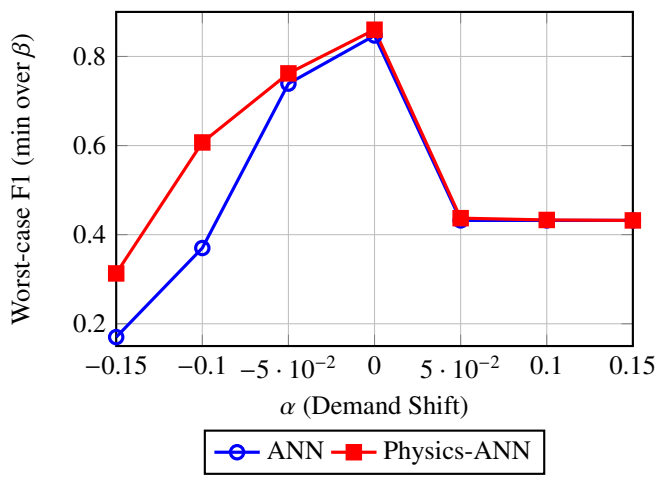

Figure 13: Worst case F1 (minimum over all β) across demand perturbations.

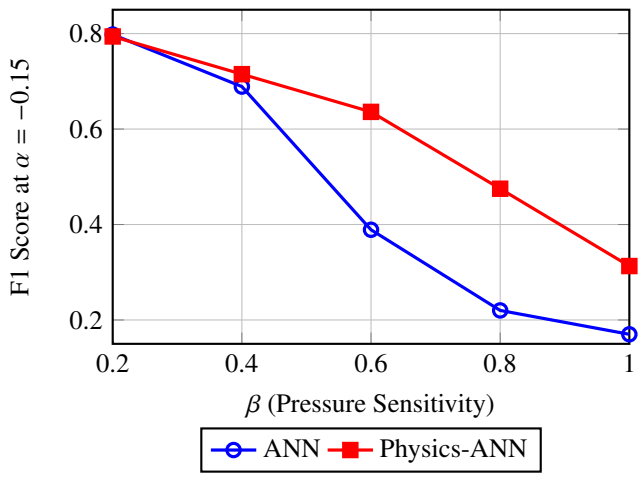

Figure 14: Performance of both models at an extreme demand shift ($\alpha = -0.15$) for varying β.

12. Limitations

Experiments were conducted on the LeakDB Hanoi benchmark, a widely used simulated dataset that allows controlled, reproducible evaluation. The Hanoi network (32 nodes, 34 pipes) serves as a standard testbed in the leak-detection literature. Training leveraged cloud compute for efficiency. The sensitivity analysis employs a tractable pressure–demand coupling to explore robustness under demand shifts; future work may incorporate full hydraulic simulation for finer-grained perturbation modeling.

13. Conclusion

We presented a Physics-Aware Artificial Neural Network (PAANN) for leak detection in Water Distribution Networks, embedding conservation of mass and the Hazen-Williams relation into the training objective. Experiments on the LeakDB Hanoi dataset show that the Physics-Aware ANN outperforms both a base ANN and XGBoost, achieving an F1 score of 0.8598 with a 62.98% reduction in false positives compared to the base ANN. By enforcing physical consistency, the model issues leak predictions only when they are hydraulically plausible, reducing false alarms and operational costs. We also introduced a sensitivity analysis framework to assess robustness under demand perturbations and discussed why classical machine learning methods such as XGBoost are less suitable for this task. Our results demonstrate that incorporating domain physics into neural network training improves detection reliability and supports the case for physics-aware approaches in critical infrastructure monitoring.

14. Future Work

Natural extensions include applying the Physics-Aware ANN to additional LeakDB topologies (e.g., Net1) and real-world SCADA datasets, incorporating EPANET-based perturbations in the sensitivity analysis, and exploring hybrid graph–physics architectures for larger networks. The rule-based localization component can be extended and benchmarked against alternative approaches. These directions build on the physics-aware framework presented here.

15. Acknowledgements

Each member of this group has contributed equally and with the same sense of enthusiasm and ownership for this project.
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16. Appendix

17. Ablation Studies

We have presented ablations on three decisions: **The Window size**, **The temporal columns**, and **the physics-aware mechanism**. All the ablations except the physics-aware mechanism are produced on the base ANN model trained on 70 scenarios and tested on 15 scenarios. The physics-aware mechanism is ablated on the Physics-Aware model trained and tested on the same scenarios.

Window-Size

The Windowing function provides a temporal context to the model, without needing recurrent layers. Table 10 shows the detection metrics for window size = 1, 6 and 12. The metrics improve on increasing the window size, and hence a window of 12 timestamps is chosen (6 hours). It is to be noted that on further increasing the size, the increase in the metrics diminish, and hence a more longer window may not provide the improvements that are justified for an exponential increase in model parameters.

Table 10: Ablation: window size.

Window	Acc.	Prec.	Rec.	F1
$W = 1$	0.8014	0.9195	0.4640	0.6168
$W = 6$	0.8271	0.9362	0.5344	0.6804
$W = 12$	0.8323	0.9009	0.5767	0.7033

Temporal columns

Table 11 shows that the *sin_hour* and *cos_hour* columns provide an improved model with better recall, f1 and accuracy for a minute trade-off in precision.

Table 11: Ablation: temporal columns.

Setting	Acc.	Prec.	Rec.	F1
Without temporal	0.8258	0.9057	0.5518	0.6858
With temporal	0.8323	0.9009	0.5767	0.7033

Physics-Aware mechanism

How physics is integrated into the model can be categorised into 2 types:

Loss-Only: The Physics Loss is added to the classification loss, but the leak head recieves no features from the predicted flow and pressue.

Coupling: Along with the Physics Loss, the leak head recieves 5 additional numerical features (mean, std, range) derived from the flow and pressure prediction head.

Table 12 shows that the model with coupling provides an increase of 0.02 in F1 over the loss-only model.

Table 12: Ablation: physics-aware mechanism (100 scenarios).

Variant	Description	F1
Loss only	Physics loss only	0.70
Coupling	pressure/flow features in leak head	0.72

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